Paper Code- AHE4081

SET-C

GLA University

Soft Skills (Even Sem)

First Mid Term Examination (2017-18)

Course- B.Tech **Year- II**

Branch- All Branches

Instructions :

1. **There are 2 sections in the paper.**
2. **Attempt all questions. The maximum marks for the test are 20.**
3. **The total time allocated is 60 minutes.**
4. **There will be no negative marking. Also, no marks will be awarded if more than one answer is marked.**
5. **The OMR form is printed on a special paper and the information given by the candidates will be scanned. Thus, the candidates are required to fill the forms carefully. Mutilated and damaged forms will be rejected by the scanner.**
6. **The candidates should ensure that the form is not folded, crumpled or mutilated in any manner. The candidates should not mark anything anywhere except wherever required. The black marks on the margin of the OMR form are important. The candidates should not tamper it with marker.**
7. **Only black& blue pen should be used to fill forms.**
8. **Each branch is identified with a specific subject code which should be written in subject column of OMR sheet**
9. **Write your University roll number in the roll no column of OMR sheet.**

Name of Student:-.....

Branch:-.....

Section-.....

Uni. Roll No:.....

Signature of Invigilator-.....

S.N	Branch	Code
1	CS	1500
2	CE	1000
3	ECE	1300
4	EN	1400
5	EE	1100
6	ME	1200
7	MCA	8840

All The Best

Section-1 (Quantitative Aptitude)- No of Questions-20, marks-10

Q1. If the cost of a calculator worth Rs 250 is increased by Rs 100, the rate of increase is
A] 100% B] 40% C] 25% D] none

Q2. A number increased by 37.5% gives 99 the number is
A] 140 B] 61.5 C] 72 D] 48

Q3. In an election between two candidates, one got 55% of the total valid votes, 20% of the votes were invalid. If the total number of votes was 7500, the number of valid votes that the other candidate got, was:
A] 2500 B] 2700 C] 2900 D] 3100

Q4. The price of an item is increased by 20% and then decreased by 20%. The final price as compared to original price
A] 4% more B] 20% more C] 20% less D] 4% less

Q5. If A's height is 40% less than that of B, how much percent B's height is more than that of A ?
A] 66.66% B] 76.66% C] 96.66% D] 86.66%

Q6. A man gains 20% by selling an article for a certain price. If he sells it at double the price, the percentage of profit will be:
A] 40% B] 100% C] 120% D] 140%

Q7. Some articles were bought at 6 articles for Rs. 5 and sold at 5 articles for Rs. 6. Gain percent is:
A] 30% B] 33.33% C] 35% D] 44%

Q8. If the cost price of 20 articles is equal to the selling price of 25 articles, what is the % profit or loss made by the merchant?
A] 25% loss B] 25% profit C] 20% loss D] 20% profit

Q9. I sell two articles, one at 20 % profit and other at 10% loss. Find overall (both the articles considered together) profit percentage, if the selling price of both the articles are 955 Rs.
A] 2.78% B] 3.41% C] 5.76 % D] None of these

Q10. In summers, a meter scale expands by 10%. What will be the profit percentage of a cloth trader, who sells at a mark up of 20% over his purchase price?
A] 8.33% B] 9.09% C] 12.5 % D] None of these

Q11. Sum of money doubles itself in 12 years. Find the rate percentage per annum.
A] 8.33% B] 9.09% C] 12.5 % D] None of these

Q12. A shopkeeper borrows Rs. 20000 from two money lenders. For one loan he paid 12% and for other 14% per annum simple interest. After one year he paid Rs. 2560 as interest. How much did he borrow from each rate?
A] Rs. 15000, Rs. 5000 B] Rs. 9000, Rs. 11000
C] Rs. 6000, Rs. 14000 D] None of these

Q13. The simple interest on Rs. 400 for 5 years together with on Rs. 600 for 4 years comes to Rs. 132, rate being same in both the cases. Find the rate of interest.
A] 3% B] 5% C] 8% D] None of these

Q14. A person level a certain sum of money at 4% simple interest, and in 5 years the interest amount to Rs. 520 less than the sum lent. Find the sum.

A] Rs. 350 B] 1300 C] Rs. 650 D] None of these

Q15. A sum was put at SI at a certain rate for 2 years. Had it been put at 3% higher rate, it would have given Rs. 300 more. Find the principal.

A] Rs. 8000 B] Rs. 5000 C] Rs. 8000 D] None of these

Q16. A sells an article to B at 20% profit. B sells the article to C at 25% profit. If the article has cost A Rs. 144, how much did C pay in Rs. for the article?

A] 216 B] 256 C] 432 D] None of these

Q17. A shopkeeper marks up his goods by 20%. As one of his friends arrives on the shop and he decides to sell the article with no profit no loss. How much discount should he give on the article, in order to make no profit no loss?

A] 25 % B] 16.67% C] 20% D] None of these

Q18. A trader lists his articles 66.66% above the cost price and allows a discount of 25% on cash payment. His gain percent is?

A] 25% B] 16.67% C] 20% D] None of these

Q19. A man sells an article at a profit of 25%. If he had bought it at 20% less and sold it for Rs 10.50 less, he would have gained 30%. Find the cost price (in rupees) of the article.

A] 50 B] 20 C] 25 D] None of these

Q20. A trader's marked price is 30% above cost price. He allows his customers 10% discount on the marked price. What profit does he make?

A] 13% B] 17% C] 10% D] None of these

Section-2 (Reasoning Ability)- No of Questions-20, marks-10

Q21. From a point, Sahil starts walking in east direction. After walking for 15 m he takes a right turn. Now he walks for 12 m before turning to his right again. Next he walks 5 m and again turns in same direction as before. He now walks for 20 m before stopping at a point. How far is this point from the point where Sahil started?

A. $4\sqrt{10}$ m B. $3\sqrt{22}$ m C. 7 m D. $2\sqrt{41}$ m E. 12 m

Q22. If South-East become North, North- East becomes West, so on what will West become?

A. North East B. North West C. South East D. South West E. NOT

Q23. From point A, Swati started walking in south direction. She walked for 4 m and took a right turn. Next she walked 5 m and turned to her left. Next she walked for 3 m and turned to her right. Next she walked 4 m and turned to her right again. Next she walked 15 m and turned to her right again and stopped at point B after walking 7 m. Find distance AB.

A. $2\sqrt{22}$ m B. $3\sqrt{21}$ m C. $2\sqrt{19}$ m D. $4\sqrt{17}$ m E. None of these

Q24. A is east of B and west of C. H is south-west of C. B is south-east of 'X'. Who is the farthest west?

A. C B. A C. X D. B

Q25. A rat runs 20' towards East & turns to right, runs 10' & turns to right, runs 9' & again turns to left, runs 5' & then turns to left, runs 12' & finally turns to left & runs 6'. Now in which direction is the rat facing?

A. East B. West C. North D. South

Q(26/27)-A has two sons. E is the daughter of G and B is the mother of C. F, the brother of E is the son of C who is the son of A. A is grandfather of J who is not a sibling of E. B has a child named D.

Q26. How is D related to J?
A. D is father of J B. D is uncle of J C. J is son of D
D. Cannot be determined E. D is niece of J

Q27. What is the ratio of males to females in the family?
A. 1 : 1 B. 1 : 3 C. 5 : 3 D. 3 : 5 E. CND

Directions (28 – 29): Consider the following relationships and answer based on the relationships given below:
P % Q' means P is daughter of Q'
P # Q' means P is wife of Q'
P $ Q' means P is father of Q'
P @ Q' means P is brother of Q'

Q28. N @ B $ M % D indicates what relationship between N and D?
A. D is sister of N B. D is nephew of N C. D is sister in law of N D. N is wife of D E. NOT

Q29. The expression —M # Q @ D % S indicates which of the following relationship?
A. Q is father of S B. D is brother-in-law of M C. M is sister of S
D. S is mother-in-law of M E. NOT

Q30. Pointing to a man in photograph, a woman said, "His brother's father is the only son of my grandfather." How is the woman related to the man in the photograph ?
A.Sister B.Daughter C.Wife D.Mother

Directions (31-32): Read the following information carefully and answer the questions given below. All the codes given below are only in two letters format.
"Banks are digital today" is written as "Zi Li Ki Ti",
"Money transfer through banks" is written as "Di Ki Si Fi",
"Digital money easy today" is written as "Si Zi Ti Bi"
"Today we have leave" is written as "Gi Xi Vi Zi".

Q31. What is the possible code for "Easy banks today"?
A. Ki Ei Zi B. Bi Ki Zi C. Bi Ti Ui D. Fi Li Xi E. NOT

Q32. What is the code for "Transfer"?
A. Di B. Ti C. Fi D. Cannot be determined E. Either Di or Fi

Direction (Q. 33-34): In each questions below is given a group of letters followed by four combinations of digits/symbols numbers.You have to find out which of the combinations correctly represents the code based on the given coding system.

Letter	P	M	A	E	J	K	D	R	W	H	I	U	T	F
Coding	4	β	1	2	3	£	5	¥	6	★	&	7	!	0

i)If the first letter is a consonant and the last letter is a vowel, the codes of both these are to be interchanged

ii)If both the first and the last letters are consonants both these are to be coded as per the code of the last letter

iii)If the first letter is vowel and the last letter is a consonant both these are to be coded as *

Q33.EAKRF

A. 21£¥0 B. *1£¥* C. 01£¥2 D. *£10¥ E. None of these

Q34. JDHIU

A. 35★&7 B. 7★5£7 C. 75★&3 D. *5★&* E. None of these

Q35. If white is called blue, blue is called red, red is called yellow, yellow is called green, green is called black, black is called violet and violet is called orange, what would be the colour of human blood ?

A. Red B. Green C. Yellow D. Violet E. Orange

Q36. If in a certain code, TEACHER is written as VGCEJGT, how would DULLARD be written in the same code ?

A. FWMNCTF B. FWNNBTE C. FWNNCSF D. FWNNCTF E. None of these

Q37. If Anil finds that he is fourteenth from the left end and 7 from the right end, then how many boys must be added to the line such that there are 30 boys in the line?

A. 8 B. 10 C. 12 D. 14 E. None of these

Q38. In a row of 25 girls, when Neha was shifted by four places towards the left, she become 10th from the left end. What was her earlier position from the right end of the row?

A. 10th B. 11th C. 12th D. 13th E. NOT

Q39. In a row of 40 boys, when Rajesh was shifted to his left by 4 places his number from the left end of the row becomes 10. What was the number of Suresh from the right end of the row if Suresh was three places to the right of Rajesh's original position?

A. 23 B. 25 C. 26 D. 27 D. 21 E. None of these

Q40. A is shorter than B but taller than C. D is taller than A. E is shorter than C. Who amongst the following is the tallest?

A. A B. B C. D D. Either B or D E. None of these

University Roll No._____

GLA UNIVERSITY, MATHURA
B. Tech. 2nd Yr. (even Sem.), Term I Examination, 2017-18

Subject: English for Professional Purposes-II Subject Code: AHE 4083
Time: 45 Minutes **Max. Marks:** 15

Notes:

- Answer all the questions from Section A, any 3 out of 4 from Section B; Section C has internal choices.
- Answers should be to-the-point, and word limit for informative questions from Section B (approx 100 words) and Section C (150 words) should strictly be maintained.

Section A **0.5 x 6=3**

1. Which syllable in the word 'Photograph' will be stressed? Indicate the stress by underlining the syllable.
2. Consider: "**pro**gress" with stress on its first syllable. Use the word appropriately to make a sentence.
3. Count the number of syllables in '*organization' and,' autocratic'*.
4. Write two sentences that exemplify the rising tone.
5. Write the first paragraph of a cover letter (this paragraph contains only the purpose of writing the letter). You are applying for the post of Executive Engineer with Samsung India, Connaught Place, New Delhi.
6. What is the meaning of 'Reverse Chronological Order' with reference to resume writing?

Section B **2x3=6**

1. In response to your order for a pair of shoes through amazon.com, you have received the shoes one size larger than your size. Complete the following paragraph of the letter, **with at least three sentences** of adjustment written for the same purpose:
   "The size-chart on the website shows.....
   .....
   .....
   .....
   also the size chart should be corrected and replaced so that other customers do not face such issue."
2. Give lay-out of a cover letter.
3. Read the advertisement given below, published in the employment news, dated 25.01.18.

Junior Engineer
Manufacture of specialty electrical, electronics and IT products *world systems Pvt. Ltd.* Seeks Junior Engineer (60) to provide support and assistance in the, installation and start-up of new facility at its Bangalore Campus. Ideal candidate will have a B.Tech. degree and 1 yr. of experience in the relevant area. Candidates without experience may be considered for trainee position initially up to the period of 6 months. Hands on experience is preferable. Interested candidates should send their resume' to: **Mr. S. K. Rao** **Sr. Technical Head, World Systems Pvt. Ltd. Bangalore.**

Write the second paragraph of the cover letter that highlights excellent achievements in your academics.

4. Write any three main headings of your resume except the Personal Information. Detail one of them.

Section C **3x2=6**

1. a. Write any six sentences that may feature in a Business letter and use one of the following expressions in each of them (Do not repeat the expressions):

   May I request you....; It may please be noted...; Hence this should not be...; How ethical this behviour of the company....; I am writing to....; The moment I noticed the fault....

   Or

   b. Answer both of the follwoing questions:
      i. Write a career objective for your resume.
      ii. Write any two points under the heading 'Technical Exposure' for your resume.

2. Answer any one of the following questions:
   a. How is The Untouchable a scathing attack on the caste system which dehumanizes an 'untouchable' like Bakha ? Explain with the help of examples from the text. **or**
   b. Enumerate and discuss any three lessons taught by Morrie to his student Mitch. Support your answer with appropriate examples from the text.

SET-A

University Roll No._____

GLA UNIVERSITY, MATHURA

B. Tech. 2nd Year (Even Semester), First Term Examination, 2017-18

Subject: English for Professional Purposes–II | Subject Code: AHE 4083

Time: 45 Minutes | Max. Marks: 40

VERBAL APTITUDE

Fill in the blanks with the most suitable option:

1. What would you say if you _____ Amitabh Bachchan?
   A. meet B. met C. would meet
   D. will meet E. None of these
2. If I _____the job and the salary was handsome, I would take it.
   A. offer B. offered C. was offered
   D. has offered E. had offered
3. If you needed a hand, I _____ come and help you out.
   A. can B. could C. will be able to
   D. would be able to E. None of these
4. If I could be any animal, I _____ a giraffe.
   A. will be B. can be C. would be
   D. was E. will have been
5. Unless you ___ smoking, you _ serious health problems.
   A. stop/ will have B. have stopped/would have
   C. stopped/would have been
   D. had stopped/would have E. None of these
6. If he ____ his own vegetables, he _____ to buy them.
   A. grow/will not B. grow/would not
   C. grew/would not have D. had grown/ would not have
   E. None of these
7. I _____ to New Zealand if I _____ the chance.
   A. travel/will have B. will travel/have
   C. would travel/have D. would travelled/had
   E. would have travelled/had
8. They _____ their new car two weeks ago.
   A. will buy B. bought C. have bought
   D. buy E. None of these
9. He's not happy because his brother _____ his computer.
   A. doesn't use B. using C. has used
   D. will use E. None of these
10. My sister ___ hamburgers, she thinks they're bad for her.
    A. will eat B. doesn't eat C. is eating
    D. eats E. None of these
11. Please be quiet! The baby _____.
    A. sleep B. sleeps C. is sleeping
    D. has slept E. None of these
12. My brother _____ football in the same club as me.
    A. is playing B. plays C. played
    D. play E. None of these

Read the given passage and fill in the blanks with the most suitable option:

I've made 'book-giving' a sort of personal crusade. I don't simply give to friends who read with hunger, I give them to everyone. I give people books _____ **(13)** the hope that they will discover the pleasure of reading. I strive _____**(14)** make it something you will enjoy.

You pick a day, ask a friend, drive there, enter the cool, silent space, run your hands over spines, settle in for an hour or two, nodding to old friends and making some new ones _____**(15)** the way. Perhaps you find my crusade extreme, but let me spell out Delhi's literary losses _____ **(16)** the last few years: It started with the closing of The Bookworm in Connaught Place and Yodakin in Hauz Khas Village. Then, last year, it was Fact & Fiction, Spell & Bound, Timeless Arts Book Studio and ED Galgotia & Sons. You can't blame _____ **(17)** me for feeling it is the end of an era.

A. for B. in C. on
D. at E. No preposition required

13. A. for B. of C. on
    D. to E. No Preposition required
14. A. across B. in C. along
    D. at E. No Preposition required
15. A. since B. over C. for
    D. of E. No Preposition required
16. A. at B. for C. on
    D. to E. No Preposition required

Identify the most preferable sentence in respect to grammar/ meaning and usage, suitable for formal writing.

17. A. The Rocky Mountains, on the border of the provinces of Alberta and British Columbia, has many attractions for tourists.
    B. This high point in a continent's water system is called a "hydrological apex", and the Columbia Icefield is one of only two such apexes in the world.
    C. Many people go to the icefield because they are the biggest area of ice and snow in North America south of the Arctic Circle.
    D. The melt water flow down these rivers into three different oceans, the Pacific, Arctic and Atlantic.
    E. Another unusual fact about the icefield are that it is one of the few place that is known for its soothing environment.
18. A. Each of us are bringing a small gift that cost less than Rs 100 for another person.
    B. Mukesh, like many other students, have been trying to decide on the best gift to buy.
    C. Mili doesn't want to get something serious; she like to make people laugh.
    D. Unfortunately, all the funny things is too expensive.
    E. Many of my classmates are arriving early to decorate the party room.
19. A. This huge field of ancient ice cover 125 square kms.
    B. The Athabasca, Stutfield and Dome glaciers makes up the icefield.
    C. As the ice melts, the water flow into four major river systems.
    D. Tourists visit the icefield from April to October each year.
    E. Walking tours and snowcoach tours is available.
20. A. Calcium, which is one of the world's most common elements, are important for growing strong bones and teeth.
    B. As far back as Ancient Egyptian times, historians tells us of the use of calcium as a building material.
    C. In nature, this important element is always found as a compound such as calcium carbonate and calcium fluoride.
    D. A person with osteoporosis can breaks a bone doing simple daily activities like walking, hugging or coughing.
    E. All sentences given above are correct.
21. A. Calcium are also used for many other purposes.
    B. Calcium-deficient bones is brittle and breaks easily.
    C. Older people, especially older women, frequently gets osteoporosis.
    D. Osteoporosis are a condition which cause calcium deficiency in bones.
    E. Every animal requires calcium for its health.

Identify the part that contains error.

22. Owing to his reputation (A)/ for being a loose cannon, (B)/ he was not included (C)/in the team working on the mega project of the company(D)/ No Error (E)
23. A large number of people (a)/ have fallen victim of (B)/ fraud companies which trick them of (C)/ their money and breach the trust (D)/ No Error (E)
24. He climbed the ladders of success (A)/ by the dint of hard work (B)/ and earned a reputation of (C) being a self made person (D)/ No Error (E)
25. These days, it's almost impossible to (A)/ pick up a newspaper or turn on the TV (B)/ without being told, of gory detail, (C)/ about the dark side of the Bitcoin story (D)/ No Error (E)
26. Caste, cash and community must not be allowed (A)/ to play any role in electoral politics (B)/and people must elect their

public representatives(C)/ on the basis of character, caliber, capacity and conduct(D)/ No Error (E)

Choose the option that explains the meaning of the idiom.

27. Every bad thing has something good or compensating.
    A. A blessing in disguise B. Once in a blue moon
    C. The buck stops here D.To get off on the wrong foot
    E. Every cloud has a silver lining
28. Something that first seems negative but turns out to be positive in the end.
    A. A blessing in disguise B. Once in a blue moon
    C. The buck stops here D. Every cloud has a silver lining
    E. To get off on the wrong foot
29. It's wasted.
    A. Know (or learn) the ropes B. He's a loose cannon
    C. To lose it D. Down the drain
    E. It's mumbo jumbo
30. A reaction without thinking.
    A. At the drop of a hat B. Knee jerk reaction
    C. Over my dead body D. It's over the top
    E. To lose it
31. Very rarely.
    A. Blue collar job B. Once in a blue moon
    C. Out of the blue D. Beat someone black and blue
    E. None of the above

Read the passages and answer the questions that follow:

It all started at the beginning of fifth grade. At first, Carmen wasn't really sure what was happening. In class, she had to squint to see the blackboard clearly. She had to do the same thing when she read street signs, or when she watched a movie. As the fuzziness got worse, she became more and more worried. It was important for her to see the notes and homework assignments the teacher put on the board.

It wasn't long before Carmen found herself squinting all the time, but she didn't want anyone to know that she was having a problem seeing. In class, she asked for a desk that was closer to the blackboard. One day, her teacher said, "Carmen, are you all right? I've noticed you squinting a lot. Are you having trouble seeing the board?"

Carmen shook her head. "I'm fine, Mrs. Cruz," she said, but she knew she couldn't pretend much longer.

At home, she had to sit closer and closer to the television in order to see the picture. Her mother noticed her squinting as she watched her favorite shows, and she began to get suspicious.

"Tomorrow I'm calling the eye doctor to set up an appointment for you," she said firmly. Carmen protested, but her mother's mind was made up.

Three days later, Carmen had new glasses and instructions from her doctor to wear them all the time. Carmen frowned in the car the whole way home. "All of the kids at school will think I'm a nerd," she said. Her mother smiled and shook her head. "You look just as beautiful with those glasses on as you do without them," she said. But Carmen didn't believe her. The next day, Carmen kept the glasses in her pocket as she walked into the schoolyard. She avoided her friends and stood alone, feeling miserable. Suddenly, she heard her friend Theresa shout. Carmen ran over to the other girls. "What's wrong?" she asked. "My silver ring is gone!" Theresa cried. "My sister sent it to me from California. It's very special and I can't lose it!"

Carmen could tell that Theresa was very upset. They all looked for the ring in the grassy area of the playground.

Carmen realized that she could search better if she could see better. She took the glasses out of her pocket and put them on. The objects and people around her came into sharp focus. She caught her breath. Everything looked so different! So clear! She looked down at the ground and a glimmer of silver caught her eye. It was the ring. "Here it is," she shouted. "I've found it!" She handed it to Theresa, and Theresa slipped the ring back on her finger.

"Thanks Carmen," she said. "I never thought we'd find it." She paused. "Hey, I didn't know you wore glasses. They look great!" Carmen had forgotten that she was wearing the new glasses. "Thanks," she replied shyly. As they walked back toward the school building, two more girls from her class complimented her glasses. Carmen smiled. "Maybe wearing glasses won't be so bad after all," she thought.

32. BEFORE Carmen got glasses she .....
    A. thought having glasses wouldn't be so bad.
    B. wasn't able to see the blackboard clearly.
    C. found Theresa's missing ring.
    D. sat far away from the television
    E. None of the above
33. Based on the end of the story, what do you think Carmen will do next?
    A. She won't tell her friends that she needs to wear glasses.
    B. She will keep her glasses in her pocket where no one can see them.
    C. She will wear her glasses all the time.
    D. She will wear her glasses only when she is with her family.
    E. None of the above
34. "Her mother noticed her squinting as she watched her favorite shows, and she began to get suspicious."
    In the sentence above, what is an antonym for the word suspicious?
    A. doubtful B. guilty C. innocent
    D. trusting E. None of the above
35. Which statement BEST describes Carmen?
    A. She is willing to overcome her fears in order to help her friends.
    B. She doesn't care how well she does in school.
    C. She cares more about herself than her friends.
    D. She doesn't worry about what other people think of her.
    E. None of the above
36. Which one of the following can be a theme of the passage?
    A. It is important to look good in the class
    B. It is important to follow the advice of the parents
    C. It is necessary to take the help of a doctor if need be
    D. Medical assistance should not be confused with the good looks
    E. Children often think more about how they look amongst their peers but parents take more practical decisions for them.

Chess is called the game of kings. It has been around for a long time. People have been playing it for over 500 years. Chess is based on an even older game from India. The chess we play today is from Europe.

Chess is a two-player game. One player uses the white pieces. The other uses the White Chess Pieces black pieces. Each piece moves in a special way. One piece is called the king. Each player has one. The players take turns moving their pieces. If a player lands on a piece, he or she takes it. The game ends when a player loses his or her king. There are a few more rules, but those are the basics.

Some people think that chess is more than a game. They think that it makes the mind stronger. Good chess players use their brains. They take their time. They think about what will happen next. These skills are useful in life and in chess. Chess is kind of like a work out for the mind.

37. What is the author's purpose in writing the second paragraph?
    A. To explain the rules of chess
    B. To compare different types of games
    C. To talk about game pieces
    D. To persuade people to play chess
    E. None of the above
38. Which is not a reason that chess is a good workout for the mind according to the text?
    A. Good chess players think about what will happen next.
    B. Good chess players take a lot of risks.
    C. Good chess players take their time.
    D. Good chess players use their brains.
    E. None of the above
40. Where did the game that chess is based on come from?
    A. Europe B. America C. India
    D. All of these E. None of these

*****

SET-B

University Roll No._____

GLA UNIVERSITY, MATHURA

B. Tech. 2nd Year (Even Semester), First Term Examination, 2017-18

Subject: English for Professional Purposes–II | Subject Code: AHE 4083

Time: 45 Minutes | Max. Marks: 40

VERBAL APTITUDE

Fill in the blanks with the most suitable option:

1. What would you do if you _____your job?
   A. lose B. lost C. would lose D. will lose E. None of these
2. If we lose the contract, there _____ be redundancies.
   A. have to B. will have to C. had to D. would have to E. None of these
3. We _____ Ravi if we'd known about his problems.
   A. help B. Helped C. had helped D. would help E. would have helped
4. Had I known the process, I _____ things differently.
   A. will do B. did C. Do D. would do E. would have done
5. What _____ do if you _____ in my position?
   A. would you/were B. will you/were C. do you/were D. will you/will be E. None of these
6. She _____ so worried about the operation if the doctor _____ things better.
   A. would not have /had explained
   B. will be/explained
   C. would not have been/had explained
   D. was/explained E. have been/ explain
7. Iron _____, if it _____wet.
   A. will rust/ will get B. rust/will get C. rusted/would got D. rust/get E. rusts/gets
8. I'm very happy: my favourite singer ___ to my town next October for a big concert.
   A. isn't going to come B. is going to come C. has come D. doesn't come E. None of these
9. Sam ___ the marathon for the first time in 2009.
   A. has run B. ran C. runs D. is running E. None of these
10. She is shocked because she ___ a terrible accident.
   A. is going to see B. has seen C. sees D. saw E. None of these
11. For dinner last night we ____ fish and chips.
   A. will have B. had C. are having D. have had E. None of these
12. Two children and one adult ____ in a fire last night.
   A. are dying B. have died C. died D. will die E. None of these

Read the passage and fill in the blanks with suitable option:

The most common excuse I hear _____ **(13)** not reading is: "I would if I had more time." 'So what happens to them?' I asked, waving my hand_____ **(14)** the books. Everyone in the room shrugged. These are people who love books. They read them and own them and understand them. We all sat _____ **(15)** silence then someone poured a little wine. The next time I visited her I stopped dead in my tracks when I saw the bookshelves. "Did you rearrange the books _____ **(16)** colour?' I asked.'You noticed!' she exclaimed.

I wonder if words will eventually be lost; whole words, paragraphs, texts-gone forever _____ **(17)** collective memory. Words are given life only by being read. Else, books become nothing but sacrificed forests where ink makes its final grave.

13. A. about B. For C. against D. at E. No Preposition required
14. A. to B. by C. n D. at E. No Preposition required
15. A. amidst B. around C. in D. of E. No Preposition required
16. A. in B. by C. from D. of E. No Preposition required
17. A. from B. into C. in D. to E. No Preposition required

Decide which sentence is the most preferable in respect to grammar/ meaning and usage, suitable for formal writing.

18. A. In Canada, one in four women over the age of fifty suffer from osteoporosis.
   B. Each osteoporosis sufferer needs to eat foods high in calcium daily.
   C. Milk, tofu and spinach is high in calcium. Regular exercise also help prevent osteoporosis.
   D. The other group that need lots of calcium are children who are growing.
   E. They need calcium to help their bones and teeth grows strong.
19. A. My favourite history professor, as well as some of his administrative staff, were asked to resign because fewer students were taking history courses than ever before.
   B. Every summer in Juhu the sound of children yelling at the beach on a weekday indicate that it is summer vaction.
   C. The selfie, which is a photograph taken by many narcissistic people, is extremely popular on various social networking websites.
   D. An effective and helpful tutor demonstrates exceptional knowledge, is responsive to the needs of her students, and listen to them carefully.
   E. Extremely careful analysis of handwriting samples show that creative people are more likely to write rounded letters.
20. A. Under my bed exists a pen and a pencil.
   B. After losing by thirty points, the team have decided to quit the game.
   C. Each of the members of the team are athletic.
   D. Every person in each of my classes are not genius.
   E. Remembering the names of all of Shakespeare's dramas is not easy.
21. A. Playing sports such as basketball, football or baseball allow children to learn teamwork.
   B. My math teacher, who is extremely demanding and yells at students, gives too much homework.
   C. Changes for the new and improved GATE is going to be implemented soon.
   D. On his forehead resides a few unsightly pimples.
   E. Eating hot momos is a fun, but less fun is its consequences.
22. A. A pair of supportive shoes are important if you are experiencing lower back pain.
   B. The paralegal believed that there were an overload of cases in the office, which was creating stress.
   C. Abhay, Sujay and Veena were incredible in each of their performances and the audience rewarded them through cheers and applause.
   D. The best part of running with bulls are the expressions on the other peoples' faces as they try to stay one step ahead of the beasts behind them.
   E. Your preference for expensive chocolates are not the reflections of what you really enjoy, but simply a manifestation of your tendency to spend money carelessly.

Identify the part that contains error.

23. The affiliation process of several new schools (A)/ too the Uttar Pradesh Board has been delayed (B)/ due to non

submission of report by(C)/ the District inspector of schools of different states.(D)/ No Error (E)

24. Besides scarcity (A)/ the price of coarse sand(B)/ has shot up (C)/ to rupees 120 per cubic feet. (D)/ No Error (E)

25. The issue is related to an erroneous atmosphere (A)/created by exaggerated importance being given to (B)/ illegal mining as also being projected and exploited by (C)/some activists for personal gains. (D)/ No Error (E)

26. In the direction of the district magistrate,(A)/ they wrote a to (B)/ the CM of the state requesting (C)/ financial support for the family. (D)/ No Error (E)

27. The district jails in the state are overcrowded by (A)/ prisoners and no effective measures (B)/ are being taken (C)/ to improve the infrastructure.(D)/ No Error (E)

Choose the correct option that explains the meaning of the idiom.

28. Go directly to the point
A. Back to square one B. Cut to the chase
C. Hit the nail on the head D. The ins and outs
E. The ball is in your court

29. Not being direct
A. Cut to the Chase B. Back to the drawing board C. Go the extra mile
D. Once in a blue moon E. Beating around the Bush

30. It's unlikely to succeed
A. Cross our fingers B. To get off on the wrong foot
C. The jury's out D. It's a long shot
E. Knee jerk reaction

31. We are very short of time
A. Working against the clock B. Hot off the press
C. To be off the hook D. Off the record
E. In the nick of time

32. To escape without punishment for your wrong
A. A drop in the ocean B. At the drop of a hat
C. Go the extra mile D. To be fishy
E. Scot-free

Read the passages and answer the questions that follow:

Primitive man was probably more concerned with fire as a source of warmth and as a means of cooking food than as a source of light. Before he discovered less laborious ways of making fire, he had to preserve it, and whenever he went on a journey he carried a firebrand with him. His discovery that the firebrand, from which the torch may very well have developed, could be used for illumination was probably incidental to the primary purpose of preserving a flame.

Lamps, too, probably developed by accident. Early man may have had his first conception of a lamp while watching a twig or fibre burning in the molten fat dropped from a roasting carcass. All he had to do was to fashion a vessel to contain fat and float a lighted reed in it. Such lamps, which are made of hollowed stones or sea shells, have persisted in identical from up to quite recent times.

The journey of fire from, first, a source of warmth to an array of other areas of life has been interesting. The discovery of coal gave a new life to fire. It made it much easier to be preserved and transported. In last century the fire was used to run the trains and the engines of the other types.

The use of fire to save the humans from the animals started only a little later than the discovery of fire. The concept of bombs must have come only after seeing the destruction that the power of fire can do.

33. Primitive man's most important use for fire was _____.
A. to provide warmth B. to cook food
C. to provide light D. Both A and B
E. None

34. The firebrand was used to _____.
A. prevent accidents B. provide light
C. scare animals D. save labour
E. None

35. By 'primary' the author means-
A. primitive B. fundamental C. elemental
D. essential E. None

36. Lamps probably developed through mere _____.
A. hazard B. fate C. chance
D. planning E. None

37. What may be an apt theme of the passage?
A. The journey of fire from primitive times to modern times
B. The power of fire C. Fire: A boon or a curse.
D. Essentials of life E. Man and his World

E.T. the Extra-Terrestrial is a video game that came out for the Atari 2600 game system in 1982. It was based on a very popular film of the same name. It cost over 125 million dollars to make. Star programmer Howard Scott Warshaw created it with consultation from Steven Spielberg. And it is widely considered to be one of the worst video games ever created. The massive failure of E.T. and its effects on Atari is an often-mentioned reason for the video game industry crash of 1983.'

It was July 27, 1982. Howard Scott Warshaw was hot off the success of his most recent game, Raiders of the Lost Ark. He received a call from Atari C.E.O. Ray Kassar. Atari had bought the rights to make a video game version of Spielberg's movie, E.T. the Extra-Terrestrial, which had just been released in June. Kassar told Warshaw that Spielberg had specifically asked for Warshaw to make the game. Warshaw was honored, but there was one huge problem. Atari needed the game finished by 1st September in order to start selling it during the Christmas season.

It had taken Warshaw six months to create Raiders of the Lost Ark. The game he made prior to that took him seven months. He was expected to create E.T. in around five weeks. Warshaw just did not have enough time to program the game properly, but he accepted the challenge anyway and production began. Spielberg wanted Warshaw to create a simple maze game, similar to Pac-Man, but Warshaw had a bigger vision. He wanted players to explore different environments in a 3D world. Warshaw followed his vision.

Atari anticipated that the game would be a huge success. Usually companies like Atari have people test games before releasing them. If there is something that testers really dislike, programmers can fix it before the public gets a chance to play. Atari decided to skip testing due to time limitations. They wanted the game released during the holiday season. It was: E.T. was released in December of 1982. They made 5 million copies and they only sold 1.5 million.

38. Which of the following is not a reason for the failure of E.T. the Extra-Terrestrial?
A. The programmer was not given enough time to finish the game properly.
B. Atari did not test the game before its release.
C. Atari made too many copies of E.T. the Extra-Terrestrial.
D. The game was released on a new system that only a small number of people owned.
E. None of the above

39. Who programmed E.T. the Extra-Terrestrial?
A. Steven Spielberg B. Ray Kassar
C. Howard Scott Warshaw D. Elliot Thomas
E. None of the above

40. Why did Atari decide to skip testing E.T. the Extra-Terrestrial?
A. Testing was too expensive.
B. Spielberg would not allow testing.
C. They were in a hurry to release the game for the holiday season.
D. Testing video games was not common until after E.T. the Extra-Terrestrial was released.
E. None of the above

*****

SET-C University Roll No._____

GLA UNIVERSITY, MATHURA

B. Tech. 2nd Year (Even Semester), First Term Examination, 2017-18

Subject: English for Professional Purposes–II Subject Code: AHE 4083

Time: 45 Minutes Max. Marks: 40

VERBAL APTITUDE

Fill in the blanks with the most suitable option:

1. What would you say if you _____ Amitabh Bachchan?
   A. meet B. met C. would meet
   D. will meet E. None of these
2. If I _____the job and the salary was handsome, I would take it.
   A. offer B. offered C. was offered
   D. has offered E. had offered
3. If you needed a hand, I _____ come and help you out.
   A. can B. could C. will be able to
   D. would be able to E. None of these
4. If I could be any animal, I _____ a giraffe.
   A. will be B. can be C. would be
   D. was E. will have been
5. What _____ do if you _____ in my position?
   A. would you/were B. will you/were C. do you/were
   D. will you/will be E. None of these
6. She _____ so worried about the operation if the doctor _____ things better.
   A. would not have /had explained B. will be/explained
   C. would not have been/had explained
   D. was/explained E. have been/ explain
7. Iron _____, if it _____wet.
   A. will rust/ will get B. rust/will get C. rusted/would got D. rust/get E. rusts/gets
8. They _____ their new car two weeks ago.
   A. will buy B. bought C. have bought
   D. buy E. None of these
9. He is not happy because his brother _____ his computer.
   A. hasn't use B. using C. has used
   D. will used E. None of these
10. She is shocked because she __ a terrible accident.
   A. is going to see B. has seen C. sees
   D. saw E. None of these
11. For dinner last night we ____ fish and chips.
   A. will have B. had C. are having
   D. have had E. None of these
12. Two children and one adult ____ in a fire last night.
   A. are dying B. have died C. died
   D. will die E. None of these

Read the given passage and fill in the blanks with the most suitable option:

I've made 'book-giving' a sort of personal crusade. I don't simply give to friends who read with hunger, I give them to everyone. I give people books _____ **(13)** the hope that they will discover the pleasure of reading. I strive _____**(14)** make it something you will enjoy-magical short stories for my sister, Dalai Lama biographies for the spiritual, Japanese fiction for the quirky-but it is always a book.

I like the pilgrimage: You pick a day, ask a friend, drive there, enter the cool, silent space, run your hands over spines, settle in for an hour or two, nodding to old friends ("Oh, hello there, Lolita new edition!") and making some new ones _____**(15)** the way. Perhaps you find my crusade extreme, but let me spell out Delhi's literary losses _____ **(16)** the last few years: It started with the closing of The Bookworm in Connaught Place and Yodakin in Hauz Khas Village. Then, last year, it was Fact & Fiction, Spell & Bound, Timeless Arts Book Studio and ED Galgotia & Sons. You can't blame _____ **(17)** me for feeling it is the end of an era.

13. A. for B. in C. on
    D. at E. No preposition required
14. A. for B. of C. on
    D. to E. No Preposition required
15. A. across B. in C. along
    D. at E. No Preposition required
16. A. since B. over C. For
    D. of E. No Preposition required
17. A. at B. for C. on
    D. to E. No Preposition required

Identify the most preferable sentence in respect to grammar/ meaning and usage, suitable for formal writing.

18. A. In Canada, one in four women over the age of fifty suffer from osteoporosis.
    B. Each osteoporosis sufferer needs to eat foods high in calcium daily.
    C. Milk, tofu and spinach is high in calcium. Regular exercise also help prevent osteoporosis.
    D. The other group that need lots of calcium are children who are growing.
    E. They need calcium to help their bones and teeth grows strong.
19. A. My favourite history professor, as well as some of his administrative staff, were asked to resign because fewer students were taking history courses than ever before.
    B. Every summer in Juhu the sound of children yelling at the beach on a weekday indicate that it is summer vaction.
    C. The selfie, which is a photograph taken by many narcissistic people, is extremely popular on various social networking websites.
    D. An effective and helpful tutor demonstrates exceptional knowledge, is responsive to the needs of her students, and listen to them carefully.
    E. Extremely careful analysis of handwriting samples show that creative people are more likely to write rounded letters.
20. A. This huge field of ancient ice cover 125 square kms.
    B. The Athabasca, Stutfield and Dome glaciers makes up the icefield.
    C. As the ice melts, the water flow into four major river systems.
    D. Tourists visit the icefield from April to October each year.
    E. Walking tours and snowcoach tours is available.
21. A. Calcium, which is one of the world's most common elements, are important for growing strong bones and teeth.
    B. As far back as Ancient Egyptian times, historians tells us of the use of calcium as a building material.
    C. In nature, this important element is always found as a compound such as calcium carbonate and calcium fluoride.
    D. A person with osteoporosis can breaks a bone doing simple daily activities like walking, hugging or coughing.
    E. All sentences given above are correct.
22. A. Calcium are also used for many other purposes.
    B. Calcium-deficient bones is brittle and breaks easily.
    C. Older people, especially older women, frequently gets osteoporosis.
    D. Osteoporosis are a condition which cause calcium deficiency in bones.
    E. Every animal requires calcium for its health.

Identify the part that contains error.

23. Owing to his reputation (A)/ for being a loose cannon, (B)/ he was not included (C)/in the team working on the mega project of the company. (D)/ No Error (E)

24. A large number of people (A)/ have fallen victim of (B)/ fraud companies which trick them of (C)/ their money and breach the trust. (D)/ No Error (E)

25. The affiliation process of several new schools (A)/ too the Uttar Pradesh Board has been delayed (B)/ due to non submission of report by(C)/ the District inspector of schools of different states. (D)/ No Error (E)

26. Besides scarcity (A)/ the price of coarse sand(B)/ has shot up (C)/ to rupees 120 per cubic feet. (D)/ No Error (E)

27. The issue is related to an erroneous atmosphere (A)/created by exaggerated importance being given to (B)/ illegal mining as also being projected and exploited by (C)/some activists for personal gains. (D)/ No Error (E)

Choose the option that explains the meaning of the idiom.

28. Go directly to the point
A. Back to square one B. Cut to the chase
C. Hit the nail on the head D. The ins and outs
E. The ball is in your court

29. Not being direct
A. Cut to the Chase B. Back to the drawing board
C. Go the extra mile D. Once in a blue moon
E. Beating around the Bush

30. It's unlikely to succeed
A. Cross our fingers B. To get off on the wrong foot
C. The jury's out D. It's a long shot
E. Knee jerk reaction

31. A reaction without thinking
A. At the drop of a hat B. Knee jerk reaction
C. Over my dead body D. It's over the top
E. To lose it

32. Very rarely
A. Blue collar job B. Once in a blue moon
C. Out of the blue D. Beat someone black and blue
E. None of the above

Read the passages and answer the questions that follow:

Primitive man was probably more concerned with fire as a source of warmth and as a means of cooking food than as a source of light. Before he discovered less laborious ways of making fire, he had to preserve it, and whenever he went on a journey he carried a firebrand with him. His discovery that the firebrand, from which the torch may very well have developed, could be used for illumination was probably incidental to the primary purpose of preserving a flame.

Lamps, too, probably developed by accident. Early man may have had his first conception of a lamp while watching a twig or fibre burning in the molten fat dropped from a roasting carcass. All he had to do was to fashion a vessel to contain fat and float a lighted reed in it. Such lamps, which are made of hollowed stones or sea shells, have persisted in identical from up to quite recent times.

The journey of fire from, first, a source of warmth to an array of other areas of life has been interesting. The discovery of coal gave a new life to fire. It made it much easier to be preserved and transported. In last century the fire was used to run the trains and the engines of the other types.

The use of fire to save the humans from the animals started only a little later than the discovery of fire. The concept of bombs must have come only after seeing the destruction that the power of fire can do.

33. Primitive man's most important use for fire was _____.
A. to provide warmth B. to cook food
C. to provide light D. Both A and B
E. None

34. The firebrand was used to _____.
A. prevent accidents B. provide light
C. scare animals D. save labour
E. None

35. By 'primary' the author means
A. primitive B. fundamental C. elemental
D. essential E. None

36. Lamps probably developed through mere .....
A. hazard B. fate C. chance
D. planning E. None

37. What may be an apt theme of the passage?
A. The journey of fire from primitive times to modern times
B. The power of fire C. Fire: A boon or a curse
D. Essentials of life E. Man and his World

Chess is called the game of kings. It has been around for a long time. People have been playing it for over 500 years. Chess is based on an even older game from India. The chess we play today is from Europe.

Chess is a two-player game. One player uses the white pieces. The other uses the White Chess Pieces black pieces. Each piece moves in a special way. One piece is called the king. Each player has one. The players take turns moving their pieces. If a player lands on a piece, he or she takes it. The game ends when a player loses his or her king. There are a few more rules, but those are the basics.

Some people think that chess is more than a game. They think that it makes the mind stronger. Good chess players use their brains. They take their time. They think about what will happen next. These skills are useful in life and in chess. Chess is kind of like a work out for the mind.

38. What is the author's purpose in writing the second paragraph?
A. To explain the rules of chess
B. To compare different types of games
C. To talk about game pieces
D. To persuade people to play chess
E. None of the above

39. Which is not a reason that chess is a good workout for the mind according to the text?
A. Good chess players think about what will happen next.
B. Good chess players take a lot of risks.
C. Good chess players take their time.
D. Good chess players use their brains.
E. None of the above

40. Where did the game that chess is based on come from?
A. Europe B. America C. India
D. All of these E. None of these

*****

First Term Examination, 2017-18
Operating Systems (CSE4001)

Uni. Roll No: .....

Time: 1.5 Hrs. **Total Marks: 20**

Notes:

1. All parts of a question should be answered at one place.
2. Answer should be brief and to-the-point and be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.
4. Figures on the right-hand side margin indicate full marks.

Section A

Note: Attempt All Questions (1x5=5)

I. Define the term 'Dispatch Latency'.

II. Which scheduling policy is most suitable for time-shared operating systems?

III. List out the data fields associated with Process Control Block.

IV. Under what circumstances would a user be better of using a timesharing system rather than a single-user workstation?

V. Describe the actions taken by kernel to context switch among processes.

Section B

Note: Attempt any Three Questions (2x3=6)

I. Discuss the different multiprocessor organizations with block diagram.

II. Discuss the reentrancy of a kernel in detail.

III. What is the difference between process termination and process suspension? List out the reasons for process suspension.

CS. B.Tech Even 2017-18 IStom
IIIyear cloud computing

IV. What different thread libraries are used? Explain any one with example.

Section C

Note: Attempt any Three Questions **(3x3=9)**

I. Enlist and discuss various services provided by the Operating Systems.

II. Discuss about the evolution of Virtual Machine. Also explain how virtualization could be implemented in operating systems.

III. What are the different states a process can be in at a particular time? What factors can lead to the transition of process? Explain with the help of process state transition diagram.

IV. Consider the 3 processes, P1, P2 and P3 shown in the table.

Process	Arrival time	Burst Time
P1	0	5
P2	1	7
P3	3	4

Describe the completion order of the 3 processes as well as average waiting time under the policies FCFS and RR2 (round robin scheduling with CPU quantum of 2 time units).

CS. B.Tech

First Term Examination, 2017-18
Discrete Mathematics (CSE 4005)

Uni. Roll No: .....

Time: 1.5 Hrs. **Total Marks: 20**

Notes:

1. All parts of a question should be answered at one place.
2. Answer should be brief and to-the-point and be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.
4. Figures on the right-hand side margin indicate full marks.

Section A

Note: Attempt All Questions **(1x5=5)**

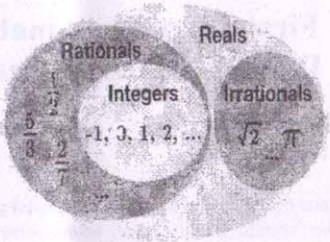
I. In Melinda's messy dresser drawer, there is a jumble of 5 red socks, 7 blue socks, 7 green socks, and 4 yellow socks. If Melinda grabs a big handful of socks without looking at what she's taking, what is the minimum number of socks Melinda has to grab in order to guarantee that she has at least 4 socks of the same color?

II. Let X={1,2,3,4,5} and Y={7,11,13} are two sets.
Consider a relation R
R={(x, y)/xϵ X and y ϵ Y and (y-x) is a perfect square}
find the ordered couples of R.

III. Find numeric function for generating function
$A(z)=2/(1-4z^2)$

IV. Is inclusion of a subset in another, i.e., ARB if and only if A is a subset of B, in the context of a universal set, an equivalence relation in the class of subsets of the universal set? Justify your answer.

V. In terms of Cardinality, are there more rational or irrational numbers or both will have the same cardinality. Justify your answers.



Section B

Note: Attempt any Three Questions **(2x3=6)**

I. Differentiate Symmetric, Asymmetric and Anti Symmetric relation with an example.

II. If set S has |S|=3 elements and satisfies
(1) $0 \in S$ and (2) if $p \in S$, $q \in S$ and $p \neq q$, then $(p + q) = S$
and 10 is an element of S, what is the other element of S excluding 0 and 10.

III. A box contains 5 different red and 6 different white balls. In how many ways can 6 balls be selected so that there are at least two balls of each type.

IV. Determine the particular solution for the following difference equations:

$$a_n - 2\,a_{n-1} + a_{n-2} = (n+1).\,2^n$$

Section C

Note: Attempt any Three Questions **(3x3=9)**

I. Let m, n and d be integers with $d \neq 0$. Then if d divides (m-n), denoted by $d \mid (m-n)$, i.e. $m-n=d\,k$ for some integer k, then we say m is congruent to n modulo d, written simply as

$m \equiv n \pmod{d}$. Let R be the relation of congruence modulo 3 on the set $\mathbf{Z}$ of all integers, i.e.

$$m\ R\ n \Leftrightarrow m \equiv n \pmod{3} \Leftrightarrow 3 \mid (m-n).$$

Show R is an equivalence relation.

II. In a class of 60 students, all the students play either football or cricket. If 20 students play both football and cricket, determine the number of players for each game if the number of students who play football is

(a) 14 more than the number of students who play cricket.

(b) exactly 5 times the number of students who play only cricket.

(c) a multiple of 2 and 3 and leaves a remainder 3 when divided by 5.

III. Let S_n be the number of comparisons needed in a Merge Sort on an array of size n. The value of S_n is represented by the following divide and conquer recurrence:

$$S_1 = 1;$$

$$S_n = 2\,S_{n/2} + n$$

Use generating function to solve S_n.

IV. A young pair of rabbits (one of each gender) is placed on an island. A pair of rabbits does not breed until they are 2 months old. After they are 2 months old, each pair of rabbits produces another pair each month. Find a recurrence relation for the number of pairs of rabbits on the island after n months, assuming that rabbits never die. Also find the solution of this recurrence relation.

FIRST TERM EXAMINATION 2017-18

Computer Networks: CSE-4003

University Roll No.....

Time: 01 ½ Hours **Total Marks: 20**

Note: Attempt **all** questions from Section-A, **any three** from Section-B and **any three** from Section-C.

Section-A

Note: Attempt all Questions (1X5=5)

Q1. Which layers in the Internet protocol stack does a router process? Which layers does a link-layer switch process?

Q2. The attenuation of a signal is -10 dB. What is the final signal power if it was originally 5 W?

Q3. What is the significance of the twisting in twisted-pair cable?

Q4. Define random access and list three protocols in this category.

Q5. Define a DC component and its effect on digital transmission.

Section-B

Note: Attempt any three questions **(2x3=6)**

Q1. Explain the relationship between data rate and signal rate. An analog signal carries 8 bits per signal element. If 1600 signal elements are sent per second, find the bit rate.

Q2. Why does a circuit-switched network need end-to-end addressing during the setup and teardown phases? Why are no addresses needed during the data transfer phase for this type of network?

Q3. How long does it take a packet of length L to propagate over a link of distance d, propagation speed s, and transmission rate R bps? Does this delay depend on packet length? Does this delay depend on transmission rate?

CS. Cloud Computing

Q4. What is the vulnerable time in CSMA, Pure ALOHA, Slotted ALOHA? What is the minimum frame size in CSMA/CD?

Section-C

Note: Attempt any three questions **(3x3=9)**

Q1. Suppose four active nodes—nodes A, B, C and D—are competing for access to a channel using slotted ALOHA. Assume each node has an infinite number of packets to send. Each node attempts to transmit in each slot with probability p. The first slot is numbered slot 1, the second slot is numbered slot 2, and so on.

a. What is the probability that node A succeeds for the first time in slot 5?
b. What is the probability that some node (either A, B, C or D) succeeds in slot 4?
c. What is the probability that the first success occurs in slot 3?

Q2. Let's consider the operation of a learning switch in the context of a network in which 6 nodes labeled A through F are star connected into an Ethernet switch. Suppose that (i) B sends a frame to E, (ii) E replies with a frame to B, (iii) A sends a frame to B, (iv) B replies with a frame to A. The switch table is initially empty. Show the state of the switch table before and after each of these events. For each of these events, identify the link(s) on which the transmitted frame will be forwarded, and briefly justify your answers.

Q3. Explain the applications of twisted pair, coaxial cable and fiber optic cable. Also explain the advantages of fiber optic cable over twisted pair, coaxial cable.

Q4. Draw the graph of the Manchester scheme using each of the following data streams (a) 11111111 (b) 01010101 (c) 00110011

CS B.Tech.

First Term Examination, 2017-18
Introduction to Microprocessors (CSE4002)

Uni. Roll No: .....

Time: 1.5 Hrs: **Total Marks: 20**

Notes:

1. All parts of a question should be answered at one place.
2. Answer should be brief and to-the-point and be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.
4. Figures on the right-hand side margin indicate full marks.

Section A

Note: Attempt All Questions **(1x5=5)**

I. Why digital bus is important in digital system?

II. Which microprocessor architecture is based on the stored-program computer concept, where instruction data and program data are stored in the same memory?

III. In digital computer, there is a need to keep track of the current executed tasks for faster execution. Which component of processor performs this?

IV. What is assembler?

V. Write example of 2-byte instruction and 1-byte instruction in 8085 microprocessor with description.

Section B

Note: Attempt any Three Questions **(2x3=6)**

I. Explain the need to de-multiplex the bus AD_7-AD_0 in 8085 MPU.

II. The instruction LDA 2050H copies the contents of the memory locations 2050H into the accumulator. It is a 3-byte instruction with four machine cycle and 13 t-states. Indentify the fourth machine cycle and its control signal with timing diagram.

III. Write assembly language instruction for following statement: (only one instruction for each)

a) Pointer to memory location 2000H, so next instruction read the contents of memory location 2000H in register B.

b) Display the contents of accumulator at monitor which is connected with port number one.

c) Clear the accumulator

d) Load two unsigned numbers in register B and register C

IV. List of four operations commonly performed by the MPU. What MPU generates to perform such operations?

Section C

Note: Attempt any Three Questions **(3x3=9)**

I. Define architecture? Why are the program counter and stack pointer are 16 bits register? While executing a program, when the 8085 MPU completes the fetching of the machine code located at the memory address 3000H, what is the content of program counter register?

II. Why addressing modes are important in MPU? Explain any three addressing modes of 8085 microprocessor with example.

III. Explain programming model of 8085. How many instructions are in 8085 MPU? How instructions are classified in 8085 MPU? Enlist all with example.

IV. Write short notes on:

a) Instruction Format

b) Microprocessor Applications and its uses

c) Difference between MPU and Microcontroller

First Term Examination, 2017-18
Object Oriented Programming (CSE4104)

Uni. Roll No: ..... Course: B.Tech. CSE/CCV/DA IV sem.

Time: 1.5 Hrs. **Total Marks: 20**

Notes:

1. All parts of a question should be answered at one place.
2. Any missing or wrong data may be assumed suitably giving proper justification.
3. Figures on the right-hand side margin indicate full marks.

Section A

Note: Attempt All Questions **(1x5=5)**

I. What is Upcasting in java? Explain how java is robust and distributed?

II. Differentiate between class variable and instance variable with an example?

III. Explain this method declaration:
static void Mymethod(String ...s)
Discuss the need for using above declarations for methods.

IV. Identify the relationship in below statement:
a) A file is an ordinary file or a directory file
b) Sachin plays for India in certain Year.

V. Explain the output for the below code:

```
abstract class Base {
abstract public void myfunc();
public void another(){
System.out.println(" My  Method");
}}
public class Abs extends Base{
public static void main(String ag[]){
Abs a =new Abs();
a.amethod();}
public void myfunc(){
System.out.println("MYfunc");
}public void amethod()
{myfunc(); }}
```

C.S Cloud Computing

Section B

Note: Attempt any Three Questions **(2x3=6)**

I. Explain four uses of keyword static in java along with complete syntax declaration for each use?

II. Is it possible to overload static method? Justify your answer with a java program? Also design a program to demonstrate how one interface extends another interface?

III. Design an object diagram for a graphical document editor using appropriate relationship symbols. Assume a document is composed of several sheets. Each sheet contains drawing objects. Drawing objects are of two types (Text and Geometrical shapes).Geometrical shapes are of four types(Line,Circle,Square and Rectangle).

IV. Explain the modifier final and its use in java? Differentiate between abstract class and interface?

Section C

Note: Attempt any Three Questions **(3x3=9)**

I. We can't instantiate an abstract class. Then why constructors are allowed in abstract class? Explain this with a suitable java example?

II. **"super class reference can store the reference of subclass object."** Explain this statement and design a program in java to justify this statement?

III. How many types of nested classes are there in java? Explain each with an example?

IV. Explain how polymorphism is implemented in java? Also discuss how the concept of generalization to specialization is achieved in java with an example.

First Term Examination,
Even Semester 2017-18

Course: B. Tech. II Yr. **Univ. No.** ----------
Title of Paper: Environmental Studies
Paper Code: AHS 3001
Time: 1.5 Hours **Max. Marks**: 20

Section-A

Note: Attempt All Questions **1 × 5 = 5 Marks**

(I) Outline important functions of lithosphere.
(II) Distinguish autecology and synecology.
(III) Differentiate between parasitic and saprophytic consumers.
(IV) What is tropopause?
(V) Name two examples of metallic and two examples of non-metallic minerals.

Section-B

Note: Attempt Any three Questions **2 × 3 = 6 Marks**

(I) Suggest to solve problems of deforestation.
(II) Discuss the effects of overuse of mineral resources.
(III) Describe the structure of marine ecosystem.
(IV) Comment on multidisciplinary nature of environmental science.

Section-C

Note: Attempt Any Three Questions **3 × 3 = 9 Marks**

(I) Draw the vertical structure of atmosphere with temperature change and explain phenomenon occur in different layers of it.
(II) Explain the scope of environmental science.
(III) Write notes on:
(a) Fluorosis (b) Fresh water crisis
(V) What do you mean by mineral resources? How can you conserve them?

***